

LLM meta-supervisor
(Offline)

Long-term logs:

Failure modes,
Disturbances,
Performance

**LLM
reasoning
agent**

**Heuristic
code
synthesis**

**Generated
heuristic code**

**Executable
Heuristic
Supervisor**
(Online)
(1-5 minute timescale)

Runs local compiled python logic

**Dynamic set-
point
optimization**

**Process
Monitoring**

**Load compiled
local heuristics**
(No API latency)

Setpoint
updates

Sensor data

**Low level
control &
execution layer**
(Sub-second time scale)

Deterministic controller (PID)

- Sub-second control
- Deterministic execution

Process (Ball Mill Circuit)

- Real time operations
- Sensors: temp, pressure, flow, etc.

Disturbances